$$R_\infty = \lim_{N\to\infty} R_N = R_0 \lim_{N\to\infty} \frac{q_N}{p_N - 1} = R_0 \lim_{N\to\infty} \frac{\frac{1}{2\sqrt{3}}\left(\lambda^N + \lambda^{-N}\right)}{\frac{1}{2}\left(\lambda^N - \lambda^{-N}\right) - 1} = \frac{R_0}{\sqrt{3}}. \quad (30)$$

2.11 **[1.5점]**

$$\frac{R_\infty}{R_0} = \frac{1}{\sqrt{3}} \approx 0.577. \quad (31)$$

.

N 2 .

3.

N	R_N	$\frac{R_N}{R_0}$	$\varepsilon = \frac{R_\infty - R_N}{R_N}$
1	R_0	1.000	-0.423
2	$\frac{2}{3}R_0$	0.667	-0.134
3	$\frac{3}{4}R_0$	0.750	-0.038
4	$\frac{7}{12}R_0$	0.583	-0.010
5	$\frac{11}{19}R_0$	0.579	<-0.004

$N = 4$ 1% .

!

$$\infty \approx 4. \quad (32)$$

	평가 항목	배점	
1.1	(1) $x_k = A\lambda^k$	0.2	**0.2**
1.2	(3) $\begin{cases} p_k = 2p_{k-1} + 3q_{k-1} \\ q_k = p_{k-1} + 2q_{k-1} \end{cases}$	0.2	**0.4**
	(5) $\begin{cases} p_{k-1} = 2p_k - 3q_k \\ q_{k-1} = 2q_k - p_k \end{cases}$	0.2	
1.3	$p_0 = 1, \quad q_0 = 0$	0.2	**0.7**
	1 1. k | p_k | q_k 0 | 1 | 0 1 | 2 | 1 2 | 7 | 4 3 | 26 | 15 4 | 97 | 56 5 | 362 | 209	0.5	
1.4	(7) $\lambda^{-k} = p_k - q_k\sqrt{3}$	0.2	**0.2**